

USHIBUKA, Japan

WMO: 478380

Lat: 32.1978N

Lon: 130.0267E

Elev: 15

StdP: 101.15

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.2	(c) 3.3	(d) -6.0	(e) 2.3	(f) 4.9	(g) -4.8	(h) 2.5	(i) 5.7	(j) 8.5	(k) 10.7	(l) 7.4	(m) 10.7	(n) 2.2	(o) 10	(p) 0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.0	(c) 33.0	(d) 25.9	(e) 32.1	(f) 25.8	(g) 31.3	(h) 25.6	(i) 27.1	(j) 30.7	(k) 26.7	(l) 30.2	(m) 26.3	(n) 29.7	(o) 4.0	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.6	29.1	25.7	21.0	28.8	25.4	20.6	28.5	85.8	31.0	83.8	30.4	82.1	29.8	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.0	6.7	5.9	0.1	35.4	1.1	1.1	-0.7	36.2	-1.3	36.9	-2.0	37.5	-2.8	38.3	
			DB												
			WB	-2.0	27.7	1.0	0.5	-2.8	28.1	-3.4	28.4	-3.9	28.7	-4.7	29.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	8.5	9.4	12.2	16.4	20.3	23.4	27.4	28.7	26.1	21.5	16.1	10.8		
	DBStd	7.32	2.86	3.17	2.90	2.58	1.87	1.83	1.76	1.50	2.16	2.53	3.15	3.18		
	HDD10.0	149	63	44	12	0	0	0	0	0	0	0	1	29		
	HDD18.3	1134	305	249	190	68	4	0	0	0	0	4	80	233		
	CDD10.0	3234	17	29	80	192	321	402	538	581	482	357	184	54		
	CDD18.3	1177	0	0	0	9	66	152	279	322	232	103	13	0		
	CDH23.3	9906	0	0	0	11	155	709	2853	3791	1990	388	9	0		
	CDH26.7	3123	0	0	0	1	8	82	912	1522	554	44	0	0		
(14) Wind	WSAvg	2.7	2.5	2.7	2.9	2.9	2.7	2.6	2.6	2.8	2.7	2.5	2.5	2.6		
(15) Precipitation	PrecAvg	2139	79	98	134	160	186	453	319	214	201	97	104	93		
	PrecMax	3435	157	204	330	395	369	1018	926	670	449	284	215	170		
	PrecMin	1342	17	40	63	81	58	88	51	28	30	9	31	7		
	PrecStd	464	30	46	54	73	92	199	206	135	132	70	50	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	19.3	21.5	24.7	27.6	29.9	33.7	35.0	32.8	29.7	24.5	19.8		
		MCWB	15.5	16.1	16.0	17.9	20.0	24.3	26.1	25.7	25.1	23.3	19.5	15.9		
	2%	DB	16.0	17.8	19.6	23.0	26.1	28.6	32.4	33.6	31.6	27.8	23.1	18.4		
		MCWB	13.2	15.1	15.2	17.1	19.6	23.4	26.0	25.9	24.9	22.1	18.5	14.9		
	5%	DB	14.4	16.3	18.4	21.8	25.0	27.6	31.6	32.7	30.6	26.6	21.9	17.0		
		MCWB	11.0	13.5	14.6	16.7	19.1	23.0	25.9	25.9	24.4	21.1	17.9	13.3		
	10%	DB	13.0	14.7	17.1	20.7	24.0	26.6	30.7	31.9	29.6	25.5	20.7	15.7		
		MCWB	9.7	11.5	13.7	16.4	18.7	22.8	25.7	25.9	24.0	20.2	16.9	12.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	17.3	18.0	20.9	23.2	26.0	27.6	27.8	27.0	25.4	21.4	17.8		
		MCDB	17.2	18.4	19.4	22.3	25.0	28.5	31.2	31.3	30.3	27.7	22.9	18.9		
	2%	WB	13.8	15.8	16.5	19.5	21.8	25.3	27.0	27.2	26.2	23.7	19.9	15.6		
		MCDB	15.5	17.4	18.4	21.2	23.5	27.2	30.5	30.9	29.5	26.6	21.6	17.5		
	5%	WB	11.7	14.0	15.4	18.4	20.9	24.7	26.6	26.9	25.6	22.5	18.7	14.0		
		MCDB	13.8	16.1	17.8	20.4	23.0	26.4	29.8	30.6	29.0	25.5	21.0	16.3		
	10%	WB	9.9	12.0	14.2	17.3	20.1	24.0	26.2	26.5	25.0	21.2	17.5	12.6		
		MCDB	12.7	14.3	16.7	19.7	22.6	25.6	29.3	30.2	28.4	24.4	20.4	15.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.2	6.6	6.9	6.8	5.4	5.5	6.0	6.1	6.3	5.9	5.7		
		MCDBR	7.1	7.5	7.9	8.4	8.3	6.7	7.0	7.1	6.9	7.2	6.3	6.7		
	5% WB	MCWBR	5.8	6.5	5.5	4.7	4.1	3.0	2.7	2.6	2.9	3.6	4.2	5.1		
		MCWBR	6.7	6.8	6.7	6.0	5.8	4.2	5.9	6.0	5.8	5.7	5.2	5.9		
(40) Clear-Sky Solar Irradiance	taub		0.437	0.472	0.527	0.546	0.555	0.560	0.542	0.523	0.502	0.463	0.432	0.427		
		taud	2.069	2.008	1.876	1.886	1.930	2.010	2.098	2.152	2.161	2.177	2.178	2.103		
	Ebn at Noon		752	768	756	763	759	753	764	773	772	771	756	737		
		Edh at Noon	134	157	193	199	191	176	161	150	144	131	120	124		
(44) All-Sky Solar Radiation	RadAvg		2.20	2.94	4.02	4.81	5.07	4.22	5.28	5.43	4.45	3.69	2.65	2.08		
	RadStd		0.23	0.29	0.26	0.43	0.57	0.69	0.67	0.43	0.46	0.38	0.26	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page